Matemàtiques
3r ESO

SOLUCIONS DE LA FITXA · UNITAT 11

Cossos geomètrics

FentMates

Intenta resoldre cada exercici abans de mirar-ne la solució. Si no t'ix el mateix resultat, repassa els passos o consulta els apunts interactius de la unitat: hi trobaràs cossos que pots girar i eines que calculen àrees i volums pas a pas. Els resultats aproximats estan arrodonits a les centèsimes amb la tecla π de la calculadora.

1 Poliedres. Fórmula d'Euler. Poliedres regulars

1. ★☆☆

Poliedre	C	V	A	$C+V=A+2$
prisma triangular	5	6	9	$11=11$
piràmide pentagonal	6	6	10	$12=12$
prisma hexagonal	8	12	18	$20=20$
piràmide octogonal	9	9	16	$18=18$
cub	6	8	12	$14=14$

2. ★☆☆

a) $8+12=A+2 \Rightarrow \boxed{A=18}$ **b)** $C+10=17 \Rightarrow \boxed{C=7}$ **c)** $12+V=32 \Rightarrow \boxed{V=20}$

3. ★☆☆

a) sí **b)** no (rectangles diferents) **c)** sí

d) no (cares de dos tipus) **e)** sí **f)** sí

Un poliedre regular té totes les cares polígons regulars iguals i el mateix nombre de cares a cada vèrtex.

4. ★★☆ $3n=24 \Rightarrow n=8$: base octogonal; $\boxed{C=10}$, $\boxed{V=16}$ $(10+16=24+2)$.

5. ★★☆ $n+1=11 \Rightarrow n=10$: base decagonal; $\boxed{C=11}$, $\boxed{A=20}$ $(11+11=20+2)$.

6. ★★☆ a) $7+7=14=12+2$: sí (per exemple, una piràmide hexagonal). b) $5+6=11\neq 12$: no.

7. ★★★ Quadrangulars: $\boxed{C=8, V=6, A=12}$ (és un octaedre si les cares són equilàters). Pentagonals: $\boxed{C=10, V=7, A=15}$. En tots dos casos $C+V=A+2$.

8. ★★★ Cada angle d'un hexàgon regular fa $120°$. En un vèrtex s'hi han d'ajuntar com a mínim 3 cares, i $3\cdot 120°=360°$: les cares quedarien planes i no tancarien cap cos.

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2 Prismes: elements i àrea

9. ★☆☆

a) $6 \cdot 4^2 = \boxed{96 \text{ cm}^2}$

b) $2(6 \cdot 3 + 6 \cdot 4 + 3 \cdot 4) = 2 \cdot 54 = \boxed{108 \text{ cm}^2}$

10. ★☆☆ $D = \sqrt{144 + 16 + 9} = \sqrt{169} = \boxed{13 \text{ cm}}$.

11. ★☆☆ $A_L = 20 \cdot 12 = \boxed{240 \text{ cm}^2}$; $A_T = 240 + 2 \cdot 25 = \boxed{290 \text{ cm}^2}$.

12. ★★☆ $A_L = 12 \cdot 10 = 120$; altura de la base $\sqrt{16 - 4} = \sqrt{12} \approx 3{,}46$; $A_{\text{base}} = \frac{4 \cdot \sqrt{12}}{2} \approx 6{,}93$. $A_T \approx 120 + 2 \cdot 6{,}93 = \boxed{133{,}86 \text{ cm}^2}$.

13. ★★☆ $A_L = 36 \cdot 8 = 288$; $a_b = \sqrt{36 - 9} = \sqrt{27} \approx 5{,}20$; $A_{\text{base}} = \frac{36 \cdot \sqrt{27}}{2} \approx 93{,}53$. $A_T \approx 288 + 187{,}06 = \boxed{475{,}06 \text{ cm}^2}$.

14. ★★☆ Base $40 \cdot 30 = 1200$; cares $2 \cdot 40 \cdot 20 + 2 \cdot 30 \cdot 20 = 2800$. Total: $\boxed{4000 \text{ cm}^2}$.

15. ★★★ $D^2 = 3l^2 \Rightarrow 3l^2 = 48 \Rightarrow \boxed{l = 4 \text{ cm}}$; $A_T = 6 \cdot 16 = \boxed{96 \text{ cm}^2}$.

16. ★★★ Hipotenusa: $\sqrt{36 + 64} = 10$. $A_L = (6 + 8 + 10) \cdot 15 = 360$; $A_{\text{base}} = \frac{6 \cdot 8}{2} = 24$. $A_T = 360 + 48 = \boxed{408 \text{ cm}^2}$.

3 Piràmides i troncs de piràmide

17. ★☆☆ $a_b = 5$; $a_p = \sqrt{144 + 25} = \boxed{13 \text{ cm}}$. $A_T = \frac{40 \cdot 13}{2} + 100 = \boxed{360 \text{ cm}^2}$.

18. ★☆☆ $A_L = \frac{24 \cdot 9}{2} = \boxed{108 \text{ cm}^2}$.

19. ★★☆ $h = \sqrt{5^2 - 3^2} = \boxed{4 \text{ cm}}$; $A_T = \frac{24 \cdot 5}{2} + 36 = \boxed{96 \text{ cm}^2}$.

20. ★★☆ $a_p = \sqrt{36 + 16} = \sqrt{52} \approx \boxed{7{,}21 \text{ cm}}$; aresta: $\sqrt{a_p^2 + 4^2} = \sqrt{68} \approx \boxed{8{,}25 \text{ cm}}$; $A_T = \frac{32 \cdot \sqrt{52}}{2} + 64 \approx \boxed{179{,}38 \text{ cm}^2}$.

21. ★★☆ Cada cara és un triangle equilàter d'altura $\sqrt{36 - 9} = \sqrt{27} \approx 5{,}20$: $A = 4 \cdot \frac{6 \cdot \sqrt{27}}{2} \approx \boxed{62{,}35 \text{ cm}^2}$.

22. ★★★ $a_p = \sqrt{4^2 + 3^2} = \boxed{5 \text{ cm}}$ (catets: l'altura i $\frac{12 - 6}{2} = 3$). $A_T = \frac{(48 + 24) \cdot 5}{2} + 144 + 36 = \boxed{360 \text{ cm}^2}$.

23. ★★★ $a_p = \sqrt{146^2 + 115^2} = \sqrt{34\,541} \approx \boxed{185{,}85 \text{ m}}$; $A_L = \frac{920 \cdot 185{,}85}{2} \approx \boxed{85\,492 \text{ m}^2}$.

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4 Cilindre i con

24. ★☆☆ $A_L = 2\pi \cdot 3 \cdot 10 = 60\pi \approx \boxed{188{,}50\ \text{cm}^2}$; $A_T = 60\pi + 18\pi = 78\pi \approx \boxed{245{,}04\ \text{cm}^2}$.

25. ★☆☆ $g = \sqrt{144 + 25} = \boxed{13\ \text{cm}}$; $A_L = \pi \cdot 5 \cdot 13 = 65\pi \approx \boxed{204{,}20\ \text{cm}^2}$; $A_T = 65\pi + 25\pi = 90\pi \approx \boxed{282{,}74\ \text{cm}^2}$.

26. ★☆☆ $h = \sqrt{100 - 36} = \boxed{8\ \text{cm}}$; $A_T = 60\pi + 36\pi = 96\pi \approx \boxed{301{,}59\ \text{cm}^2}$.

27. ★★☆ $r = 4$: $A_T = 2\pi \cdot 4 \cdot 12 + 2\pi \cdot 16 = 96\pi + 32\pi = 128\pi \approx \boxed{402{,}12\ \text{cm}^2}$.

28. ★★☆ $g = \sqrt{81 + 144} = 15$; només la superfície lateral: $A_L = \pi \cdot 9 \cdot 15 = 135\pi \approx \boxed{424{,}12\ \text{cm}^2}$.

29. ★★☆ $2\pi r \cdot 10 = 120\pi \Rightarrow \boxed{r = 6\ \text{cm}}$; $A_T = 120\pi + 72\pi = 192\pi \approx \boxed{603{,}19\ \text{cm}^2}$.

30. ★★★ L'arc del sector fa $2\pi \cdot 4 = 8\pi$ i la circumferència sencera de radi 12 faria 24π: és $\frac{8\pi}{24\pi} = \frac{1}{3}$ del cercle. Angle: $\frac{360^\circ}{3} = \boxed{120^\circ}$.

31. ★★★ $A = 12 \cdot 2\pi \cdot 0{,}4 \cdot 5 = 48\pi \approx 150{,}80\ \text{m}^2$; $\frac{150{,}80}{4} \approx \boxed{37{,}70\ \text{L}}$.

5 L'esfera. La Terra

32. ★☆☆

a) $64\pi \approx \boxed{201{,}06\ \text{cm}^2}$ **b)** $100\pi \approx \boxed{314{,}16\ \text{cm}^2}$ **c)** $\pi \approx \boxed{3{,}14\ \text{m}^2}$

33. ★★☆ $4\pi r^2 = 144\pi \Rightarrow r^2 = 36 \Rightarrow \boxed{r = 6\ \text{cm}}$.

34. ★★☆ Mitja esfera: $\frac{4\pi \cdot 25}{2} = 50\pi \approx \boxed{157{,}08\ \text{m}^2}$.

35. ★★☆ $40^\circ + 10^\circ = \boxed{50^\circ}$; $\frac{40\,000 \cdot 50}{360} \approx \boxed{5555{,}56\ \text{km}}$.

36. ★★☆ Cada 15° és una hora: 45° E $\rightarrow$ 3 hores més: $\boxed{15h}$; 30° O $\rightarrow$ 2 hores menys: $\boxed{10h}$.

37. ★★★ $\frac{4\pi \cdot 6371^2}{4\pi \cdot 1737^2} = \left(\frac{6371}{1737}\right)^2 \approx \boxed{13{,}45}$ vegades.

6 Volum. Prismes i cilindres

38. ★☆☆

a) $\boxed{3200\ \text{L}}$ **b)** $\boxed{0{,}45\ \text{L}}$ **c)** $\boxed{2500\ \text{cm}^3}$

d) $\boxed{750\ \text{mL}}$ **e)** $\boxed{12\ \text{m}^3}$ **f)** $\boxed{0{,}5\ \text{m}^3}$

39. ★☆☆

a) $6^3 = \boxed{216\ \text{cm}^3}$ **b)** $\boxed{120\ \text{cm}^3}$ **c)** $160\pi \approx \boxed{502{,}65\ \text{cm}^3}$

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40. ★☆☆ $V = \frac{5 \cdot 12}{2} \cdot 20 = \boxed{600 \text{ cm}^3}$.

41. ★★☆ $a_b = \sqrt{4-1} = \sqrt{3}$; $A_{\text{base}} = \frac{12 \cdot \sqrt{3}}{2} = 6\sqrt{3} \approx 10{,}39$; $V \approx \boxed{51{,}96 \text{ cm}^3}$.

42. ★★☆ $V = 72\,000 \text{ cm}^3 = \boxed{72 \text{ L}}$; $\frac{3}{4} \cdot 72 = \boxed{54 \text{ L}}$.

43. ★★☆ $V = \pi \cdot 1{,}5^2 \cdot 4 = 9\pi \approx 28{,}27 \text{ m}^3 \approx \boxed{28\,274 \text{ L}}$.

44. ★★☆ Base (el lateral de la piscina): $\frac{(1+3) \cdot 25}{2} = 50 \text{ m}^2$; $V = 50 \cdot 10 = \boxed{500 \text{ m}^3}$.

45. ★★★ $\pi \cdot 16 \cdot h = 500 \Rightarrow h = \frac{500}{16\pi} \approx \boxed{9{,}95 \text{ cm}}$.

46. ★★★ $V = 24 \cdot 11{,}5 \cdot 5 - 3 \cdot \pi \cdot 1{,}5^2 \cdot 5 = 1380 - 33{,}75\pi \approx \boxed{1273{,}97 \text{ cm}^3}$.

7 Volum de piràmides, cons i esferes

47. ★☆☆

a) $\frac{36 \cdot 10}{3} = \boxed{120 \text{ cm}^3}$ **b)** $21\pi \approx \boxed{65{,}97 \text{ cm}^3}$ **c)** $288\pi \approx \boxed{904{,}78 \text{ cm}^3}$

48. ★☆☆ $r = 10$: $V = \frac{4}{3}\pi \cdot 1000 \approx \boxed{4188{,}79 \text{ cm}^3} \approx \boxed{4{,}19 \text{ L}}$.

49. ★★☆ $h = \sqrt{13^2 - 5^2} = \boxed{12 \text{ cm}}$; $V = \frac{100 \cdot 12}{3} = \boxed{400 \text{ cm}^3}$.

50. ★★☆ $h = \sqrt{100 - 36} = 8$; $V = \frac{\pi \cdot 36 \cdot 8}{3} = 96\pi \approx \boxed{301{,}59 \text{ cm}^3}$.

51. ★★☆ $V = \frac{4}{3}\pi \cdot 11^3 \approx 5575{,}28 \text{ cm}^3 \approx \boxed{5{,}58 \text{ L}}$.

52. ★★☆ $a_b = \sqrt{12}$; $A_{\text{base}} = \frac{24 \cdot \sqrt{12}}{2} \approx 41{,}57$; $V = \frac{A_{\text{base}} \cdot 9}{3} \approx \boxed{124{,}71 \text{ cm}^3}$.

53. ★★★ $V = \frac{1}{2} \cdot \frac{4}{3}\pi \cdot 12^3 = 1152\pi \approx 3619{,}11 \text{ cm}^3 \approx \boxed{3{,}62 \text{ L}}$.

54. ★★★ $\frac{1}{3}\pi \cdot 36 \cdot h = \frac{4}{3}\pi \cdot 216 \Rightarrow 12h = 288 \Rightarrow \boxed{h = 24 \text{ cm}}$ (quatre vegades el radi).

8 Cossos compostos i problemes

55. ★☆☆ $V = 6^3 + \frac{36 \cdot 4}{3} = 216 + 48 = \boxed{264 \text{ cm}^3}$.

56. ★★☆ $V = \pi \cdot 9 \cdot 10 + \frac{\pi \cdot 9 \cdot 4}{3} = 90\pi + 12\pi = 102\pi \approx \boxed{320{,}44 \text{ m}^3}$.

57. ★★☆ $V = \pi \cdot 4 \cdot 6 + \frac{4}{3}\pi \cdot 8 = 24\pi + \frac{32\pi}{3} = \frac{104\pi}{3} \approx \boxed{108{,}91 \text{ cm}^3}$; $A = 2\pi \cdot 2 \cdot 6 + 4\pi \cdot 4 = 40\pi \approx \boxed{125{,}66 \text{ cm}^2}$.

58. ★★☆ $V = \frac{\pi \cdot 2{,}5^2 \cdot 12}{3} + \frac{1}{2} \cdot \frac{4}{3}\pi \cdot 2{,}5^3 = 25\pi + \frac{125\pi}{12} = \frac{425\pi}{12} \approx \boxed{111{,}26 \text{ cm}^3}$.

59. ★★☆ $V = \pi(6^2 - 5^2) \cdot 50 = 550\pi \approx \boxed{1727{,}88 \text{ cm}^3}$.

60. ★★☆ $V = 1000 - \pi \cdot 9 \cdot 10 = 1000 - 90\pi \approx \boxed{717{,}26 \text{ cm}^3}$.

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61. ★★★ Si el con menut fa x d'altura: $\frac{x}{12} = \frac{x+24}{18} \Rightarrow x = 48$. $V = \frac{\pi \cdot 18^2 \cdot 72}{3} - \frac{\pi \cdot 12^2 \cdot 48}{3} = 7776\pi - 2304\pi = 5472\pi \approx 17\,190{,}80$ cm^3 $\approx \boxed{17{,}19 \text{ L}}$.

62. ★★★ El volum de la bola és $\frac{4}{3}\pi \cdot 27 = 36\pi$. $\pi \cdot 36 \cdot h = 36\pi \Rightarrow \boxed{h = 1 \text{ cm}}$.

63. ★★★ $V = \pi \cdot 4 \cdot 5 + \frac{1}{2} \cdot \frac{4}{3}\pi \cdot 8 = 20\pi + \frac{16\pi}{3} = \frac{76\pi}{3} \approx \boxed{79{,}59 \text{ m}^3}$. $\frac{79\,587}{50} \approx 1591{,}7$ min $\approx \boxed{26{,}5 \text{ hores}}$.

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